

1.3 Components of HCI

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد...

CS XII

Chapter 1: Computer Systems (HCI)

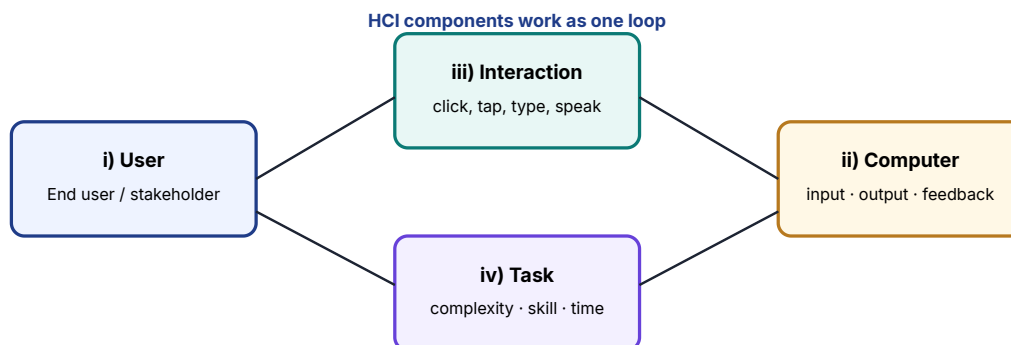
Lecture 04 of 16

1.3 Components of HCI

LEARN IT

HCI relies on several primary components that work together. If one component is weak, the whole interaction feels broken. The first four are the people, the machine, the conversation, and the job to be done.

	Component	Meaning
i	User (صارف)	Any individual or group that interacts with a computer system to achieve defined goals. Two categories: end users (people who directly operate the system, from beginners to experts) and stakeholders (people with an interest in the outcome, such as managers and clients).
ii	Computer system (کمپیوٹر سسٹم)	The platform where humans and computers communicate through input, output and feedback using an interface. It connects human actions with machine responses.
iii	User interaction (صارف کا تعامل)	The process through which a user communicates with a system by performing actions (clicking, tapping, typing, speaking). It transforms user intent into system responses.
iv	Task (کام)	The specific action users perform (for example, booking a ticket). It has three characteristics: task complexity (filling a form vs designing a 3D model), user knowledge (basic typing vs advanced programming), and time requirement (ATM withdrawal vs online passport application).



The user performs a task through interaction with a computer system. All four parts must fit together.

WORKED EXAMPLE

Worked example: label the four components of an HBL ATM withdrawal.

1. **User:** end user = the account holder standing at the machine; stakeholder = the bank that owns the ATM and must keep it safe and working.
2. **Computer system:** the ATM hardware plus the bank's software; it takes PIN input, shows a screen, and dispenses notes (output and feedback).
3. **User interaction:** insert card → type PIN → choose "Cash" → type amount → confirm. Intent ("I want Rs 5,000") becomes a machine response (cash + receipt).
4. **Task:** withdraw cash. Complexity: low (a few buttons). User knowledge: basic (PIN and amounts). Time: about one minute — much shorter than an online passport application.

REMEMBER IT

U-C-I-T: User, Computer system, Interaction, Task. "U Can Interact with Tasks." Then add three task questions: How hard? What skill? How long?

COMMON MISTAKE

Mistake: Calling the bank manager an "end user" of the ATM, or saying a task has only one property (difficulty).

Fix: The manager is usually a **stakeholder**. The person pressing the keys is the **end user**. A task always has complexity, knowledge and time — mention all three in long answers.

اردو میں سمجھیں

ایچ سی آئی کے چار بنیادی اجزاء ہیں: صارف (اینڈ یوزر جو خود سسٹم چلاتا ہے اور اسٹیک ہولڈر جس کا نتیجے میں مفاد ہے)، کمپیوٹر سسٹم (ان پٹ، آؤٹ پٹ اور فیڈبیک کا پلیٹ فارم)، صارف کا تعامل (کلک، ٹیپ، ٹائپ، بولنا) اور کام (ٹاسک) جس کی پیچیدگی، مطلوبہ مہارت اور وقت مختلف ہوتا ہے۔ اے ٹی ایم سے رقم نکالنا آسان اور مختصر کام ہے؛ پاسپورٹ کی آن لائن درخواست لمبا کام ہے۔

CHECK YOURSELF

1. What is the difference between an end user and a stakeholder?
2. Name the three characteristics of a task.
3. In a school portal, who is a stakeholder?

Answers: 1. End users directly operate the system; stakeholders have an interest in the outcome (managers, clients) but may not click the buttons. 2. Task complexity, user knowledge, time requirement. 3. The principal, IT admin or board office — they care about results even if students are the end users.

Next lecture: Types of User Interaction